

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022

Course: B. Tech. Branch :Electrical Engineering Semester :VIIIth

Subject Code & Name: BTEEP801:2 DC Power Transmission System

Max Marks: 60

Date: 07/07/2022

Duration: 3.45 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Analysis of Graetz circuit with overlap.	CO1	6
B) Explain Choice of Converter Configuration	CO2	6
C) Explain advantages and disadvantages of DC transmission system and applications.	CO3	6
Q.2 Solve Any Two of the following.		12
A) Explain Capacitor commutated converter.	CO1	6
B) Explain Converter control characteristics	CO2	6
C) Explain Fourier series, analysis of 6 pulse LCC neglecting overlap	CO3	6
Q. 3 Solve the following.		12
A) Analysis of 6 pulse LCC neglecting inductance	CO1	6
B) Explain Extinction angle: Commutation margin angle for normal inverter operation of 6 pulse LCC.	CO2	6
Q.4 Solve Any Two of the following.		12
A) Explain Steady state analysis of a general LCC	CO1	6
B) Explain Types of AC Filters	CO2	6
C) Explain characteristics and Non-characteristic harmonics.	CO3	6
Q. 5 Solve the following.		12
A) Explain Principles of DC Link Control with neat diagrams and derivation.	CO1	6
B) Explain Types of MTDC systems with neat diagrams.	CO2	6

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